21-242 Recitation 12

RREF — Q1. The RREF algorithm, and (A)

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

State the RREF algorithm and compute the RREF of $A = \begin{pmatrix} 0 & 1 & 0 \\ 3 & 0 & 1 \\ 6 & 5 & 2 \end{pmatrix}$.

Course-level solution

Algorithm (Gaussian elimination to reduced row echelon form). Start with A and a "current" top-left position $(r, c) = (1, 1)$. Repeat until you run out of rows or columns:

1. Among rows $r, \ldots, m$, find a row whose entry in column c is nonzero. If none exists, increment c and repeat.
2. Swap that row with row r (a pivot now sits at (r, c)).
3. Scale row r so the pivot equals 1.
4. Add multiples of row r to every other row so that all other entries in column c become 0.
5. Increment r and c and continue.

The output is RREF(A): each pivot is 1, each pivot is the first nonzero entry of its row, and each pivot column is a standard basis vector.

Computation. Column 1 is nonzero. Swap row 1 with row 2, then scale the new row 1 by 1/3:

$$\begin{pmatrix} 3 & 0 & 1 \\ 0 & 1 & 0 \\ 6 & 5 & 2 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 1/3 \\ 0 & 1 & 0 \\ 6 & 5 & 2 \end{pmatrix}.$$

Subtract 6 times row 1 from row 3:

$$\begin{pmatrix} 1 & 0 & 1/3 \\ 0 & 1 & 0 \\ 0 & 5 & 0 \end{pmatrix}.$$

Subtract 5 times row 2 from row 3:

$$\begin{pmatrix} 1 & 0 & 1/3 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Pivots are already 1 and already clear above. So

$$\mathrm{RREF}(A) = \begin{pmatrix} 1 & 0 & 1/3 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$